

KAIMING YANG

M.Sc. Computer Engineering · Singapore

📞 (+65)80845754

✉️ ykaiming@u.nus.edu

🐙 GitHub

🌐 LinkedIn

🎓 Scholar



Focused on AI models that perceive and generate environment states, connecting multimodal understanding and generation. My work centers on Physical AI through video generation and world models, complemented by multimodal-agent research. I work on camera control and controllable video generation: training models to follow editable camera trajectories, using 3D spatial-state and consistency modeling to support action-conditioned future video prediction, and evaluating generated videos for physical plausibility and robot-manipulation executability.

Available full-time: Feb. 2027 · English: IELTS 6.5



Education

National University of Singapore QS World #10

Aug. 2025—Jan. 2027 (Expected)

M.Sc. · Computer Engineering Department of Electrical and Computer Engineering · GPA 4.5/5.0

Xidian University Project 211 Double First-Class Computer Science A- (4th Round)

Sep. 2021—Jun. 2025

B.Eng. · Software Engineering School of Computer Science and Technology · GPA 3.9/4.0; rank 9/120



Internship Experience

● Huawei Singapore Research Institute · 2012 Labs

Research Intern · Singapore

Jun. 2026—Dec. 2026

Multimodal Model Lab

Geometry-Aware World Model Action-Conditioned Geometry & Video Generation

• **Project Overview:** Studies action-conditioned causal joint generation of geometry and video, extending pixel-only world-model representations to joint geometry-and-appearance states through geometry supervision and action control.

• **Project Work:**

- **Multimodal Architecture:** Aligned 3D geometry representations to video latents through a geometry autoencoder and designed a multimodal DiT with joint attention, using geometry to supervise video generation.
- **Action-Conditioned Joint Generation:** Analyzed and validated actions' more direct guidance of geometry changes; conditioned joint geometry/video-latent generation on actions and existing geometry/video states to causally generate future geometry and video.
- **Geometry Supervision & Validation:** Validated geometry supervision for video generation, using geometric information in joint generation to guide appearance generation toward physical geometric structure and connect action control, geometry changes, and visual generation.

Technical Keywords: Physical AI World Models Representation Learning

Camera-Controlled Video Generation Camera-Control Adapter & Editable-Trajectory Evaluation

• **Project Overview:** Trains an existing camera-control Adapter across multiple video backbones and evaluates precise, compositional control with editable trajectories.

• **Project Work:**

- **Control Modeling:** Reused an existing camera-control method and trained/adapted its Adapter across multiple pretrained video backbones, enabling generation from supplied camera trajectories with distributed training, checkpoint recovery, and camera-condition comparisons.
- **Trajectory Design:** Addressed difficult-to-edit, imbalanced video trajectories with 14 directional primitive actions and compound moves spanning rotation, translation, and zoom; normalized translation by reference depth with adjustable magnitude and speed.
- **Multidimensional Evaluation:** Evaluated video generators and video world models across geometry fidelity, control response, and appearance/world consistency, examining trajectory error, direction/amplitude, compound execution, and scene preservation with control comparisons.

Technical Keywords: Controllable Generation World Models Model Evaluation

Camera-Aware Video Data Pipeline Camera-Action & Scene-Semantics Data Pipeline

• **Project Overview:** A video data-production pipeline providing traceable camera-motion and scene-semantic supervision.

• **Project Work:**

- **Action Annotation:** Used depth-aware, confidence-weighted decomposition of adjacent-frame motion from reconstructed depth/poses to yield continuous 14D rotation/translation/zoom scores and multi-label pseudo-labels for video/world-model training, motion filtering and evaluation.
- **Caption Refinement:** VLM cleaning + risk-based human QA/editing tools preserve scene semantics and remove camera-trajectory leakage.
- **Production & Quality Control:** Unified camera trajectories, intrinsics, action intensities, and pseudo-labels with geometry/action evaluation, threshold calibration, and visual failure review; preserved raw scores for relabeling without reconstruction, with provenance, multi-machine processing, and resumable runs.

Technical Keywords: Multimodal Data 3D Vision Data-Centric AI

● Yonyou Network Technology Co., Ltd.

R&D Intern · Beijing

Jul. 2023—Aug. 2023

Model Operations Department

Enterprise Dashboards & Platform Components ERP Data Integration & Reusable Components

• **Project Overview:** Enterprise ERP data integration connecting business data, API services, and reusable dashboards.

• **Project Work:**

- **Data & Components:** Integrated multi-table ERP data through joins, aggregation, and metric validation with Yonyou BIP semantic models and SQL; delivered reusable dashboard templates and visual panels, plus a date-variable component for configurable reporting periods.
- **Integration & Delivery:** Implemented authenticated API ingestion, structured-data parsing, CSV export, and request/response encryption; optimized concurrent data I/O and tested Spring Boot API requests.

Technical Keywords: Enterprise Software Data Engineering Platform Development



Research Experience

Showlab · National University of Singapore

Student Researcher

Oct. 2024—May 2026

Multimodal Agents & Generated Video Evaluation
Supervisor: [Prof. Mike Zheng Shou](#)

Dream.exe Video Physics & Robot Executability Benchmark

[Paper](#) [Code](#)

- **Project Overview:** A benchmark of physical plausibility and robot executability in generated manipulation videos.
- **Project Work:**
 - **Evaluation Design:** Led the complete evaluation pipeline, combining VLM judgments of visual/physical plausibility with robot execution after trajectory recovery to assess physical properties, action executability, and task completion.
 - **Trajectory Extraction & Execution:** Built the complete trajectory extraction and execution framework from scratch for simulated and real videos: combined end-effector/object tracking with depth estimation using a LoRA-fine-tuned model, computed robot-arm actions from point-cloud motion, and validated execution in robosuite/MuJoCo.
 - **Benchmark & Experiments:** Built 101 manipulation tasks across three levels of physical complexity and integrated/evaluated eight video and world-action models, jointly analyzing visual/physical quality, trajectory fidelity, execution success, and task completion.

Technical Keywords: [Physical AI](#) [Video Generation](#) [Embodied AI](#)

WorldGUI GUI Agent Framework & Dynamic-State Benchmark

[Paper](#) [Code](#)

- **Project Overview:** A dynamic-state benchmark and GUI agent framework for adaptive planning, execution, and error correction.
- **Project Work:**
 - **Agent Framework:** Co-designed and implemented the Plan-Act-Critic GUI Agent Framework, with post-planning, pre-action, and post-action checks enabling self-reflection and error detection; feedback revises plans and actions for bounded iterative correction and improvement.
 - **Memory & Task State:** Designed short-term memory for each subtask's execution trajectory and longer-term task state through plan-progress tracking, grounding decisions in current task and environment states.
 - **Context & Model Decisions:** Built GUI parsers and software-knowledge skills for each agent module; combined them with memory and plan progress to provide context for MLLM state understanding, planning, decisions, and action execution.
 - **Benchmark & Validation:** Co-designed tasks and augmented initial states to scale scarce GUI task data to 611 instances across 10 apps; conducted WorldGUI/WindowsAgentArena evaluations and ablations showing Critic's key role in execution and correction.
 - **Capability Evaluation:** Evaluated GUI task execution across MLLMs, GUI-VLAs such as UI-TARS, and GUI agents such as Agent-S2.

Technical Keywords: [Multimodal Agents](#) [Computer Use](#) [Planning & Feedback](#)

CriticGUI GUI Completion Judgment & Corrective Feedback

- **Project Overview:** An offline benchmark of multimodal GUI critics for task-completion judgment and corrective feedback.
- **Project Work:**
 - **Interaction Capture:** Developed a GUI recording tool that aligns mouse/keyboard events, screenshots, and video from human demonstrations and agent exploration into semantic atomic actions and state transitions.
 - **Dataset Construction:** Combined LLM pre-annotation and human review for ~540 examples across five apps (1:4 positive/negative); correct completion versus lagging, premature, partial, and erroneous execution tests task-environment state alignment.
 - **Model Evaluation:** Evaluated completion judgments with accuracy/recall and feedback similarity to human-refined references with BERTScore, examining judgment and feedback separately.

Technical Keywords: [Agent Evaluation](#) [Multimodal Understanding](#) [Human-AI Collaboration](#)

ICTT · Xidian University

Student Researcher

Oct. 2022—Oct. 2023

NLP & Formal Specification
Supervisor: [Prof. Bin Yu](#)

nl2spec++ Temporal Reasoning & Specification Generation

[Code](#)

- **Project Overview:** Cross-sentence temporal reasoning that turns natural-language requirements into formal specifications.
- **Project Work:**
 - **Relational Reasoning:** Used GPT-3.5/GPT-4 with few-shot reasoning demonstrations to compose relation chains through shared entities and infer temporal relations between entities that do not directly co-occur.
 - **Specification Generation:** Translated inferred relations into Linear Temporal Logic (LTL) formulas and validated syntax with a parser, connecting relation reasoning to specification generation.

Technical Keywords: [LLM Reasoning](#) [Natural Language Processing](#) [Formal Methods](#)



Publications

Selected Publications

Selected papers and publication status

- [Dream.exe — co-first author; Spotlight at the ICML 2026 FoGen Workshop; under review at ICLR 2027.](#)
- [WorldGUI — second author; ACL 2025 REALM Workshop.](#)
- [AdaHC — third author; accepted at ICML 2026.](#)
- [Libra — third author; arXiv preprint \(2607.23250\).](#)



Technical Skills

- **Research:** Video Generation · World Model · Robotics · 3D Geometry · Multimodal Agent · Proactive Agent
- **Programming:** Python · C/C++ · Java · LaTeX · Markdown · SQL
- **Frameworks:** LangGraph · PyTorch · Diffusers · FSDP
- **DevOps:** Git · Linux · Docker · Weights & Biases



Honors & Intellectual Property

Mathematical Contest in Modeling — Finalist [2024](#) University Scholarships 2023 / 2024 Patent [CN117312137A](#)